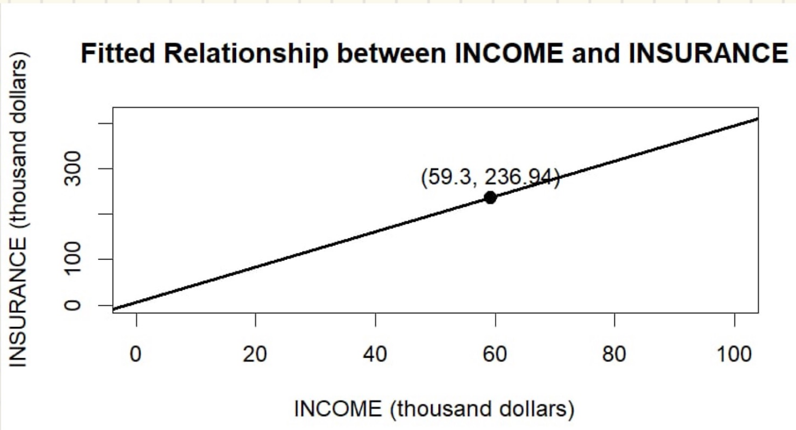
3.18 A life insurance company examines the relationship between the amount of life insurance held by a household and household income. Let *INCOME* be household income (thousands of dollars) and *INSURANCE* the amount of life insurance held (thousands of dollars). Using a random sample of $N = 20$ households, the least squares estimated relationship is

$$\widehat{INSURANCE} = \underset{(7.383)}{6.855} + \underset{(0.112)}{3.880}\,INCOME \qquad \text{(se)}$$

a. Draw a sketch of the fitted relationship identifying the estimated slope and intercept. The sample mean of $INCOME = 59.3$. What is the sample mean of the amount of insurance held? Locate the point of the means in your sketch.

b. How much do we estimate that the average amount of insurance held changes with each additional \$1000 of household income? Provide both a point estimate and a 95% interval estimate. Explain the interval estimate to a group of stockholders in the insurance company.

a)



$\widehat{insurance} = 6{,}855 + 3{,}880 \times 59{,}3 = 236{,}94$

b)

$\Delta insurance = 3{,}880 \times 1000 = \$\,3880$

→ Increase \$ 3880 insurance with each additional \$1000 of household income.

$df = 20 - 2 = 18$

$t_{0{,}975,18} = 2{,}101$

$CI = \hat{\beta}_1 \pm t_{0{,}975,18} \times se(\hat{\beta}_1)$

$= 3{,}880 \pm 2{,}101 \times 0{,}112 \Rightarrow [3{,}645 ; 4{,}115]$

In repeated sampling, about 95% intervals constructed this way will containt the true value of parameter β_1

or : With 95% confidence, each additional \$1000 income increases insurance by \$3645 to \$ 4115.

c. Construct a 99% interval estimate of the expected amount of insurance held by a household with \$100,000 income. The estimated covariance between the intercept and slope coefficient is −0.746.

d. One member of the management board claims that for every \$1000 increase in income the average amount of life insurance held will increase by \$5000. Let the algebraic model be $INSURANCE = \beta_1 + \beta_2 INCOME + e$. Test the hypothesis that the statement is true against the alternative that it is not true. State the conjecture in terms of a null and alternative hypothesis about the model parameters. Use the 5% level of significance. Do the data support the claim or not? Clearly, indicate the test statistic used and the rejection region.

e. Test the hypothesis that as income increases the amount of life insurance held increases by the same amount. That is, test the null hypothesis that the slope is one. Use as the alternative that the slope is larger than one. State the null and alternative hypotheses in terms of the model parameters. Carry out the test at the 1% level of significance. Clearly indicate the test statistic used, and the rejection region. What is your conclusion?

c. income = 100 (thousands $)

$$\widehat{\text{insurance}} = 6,855 + 3,880 \times 100 = 394,855$$

$$se(\hat{y}) = \sqrt{Var(\hat{B}_0) + x^2 Var(\hat{B}_1) + 2x\, cov(\hat{B}_0, \hat{B}_1)}$$

$$= \sqrt{7,383^2 + 100^2 . 0,112^2 + 2.100.(-0,746)}$$

$$= 5,545$$

$$t_{0,995,18} = 2,878$$

$$CI = \hat{y} \pm t_{0,995,18} \times se(\hat{y})$$

$$= 394,855 \pm 2,878 \times 5,545 \Rightarrow [378,90 ; 410,81]$$

d) Hypotheses :

$$H_0 : B_1 = 5$$

$$H_1 : B_1 \neq 5$$

Test statistic :

$$t_{stat} = \frac{b_1 - 5}{se(b_1)} = \frac{3,880 - 5}{0,112} = -10$$

Critical value

$$t_{0,975,18} = 2,101$$

$\Rightarrow |t| > 2,101$ $(10 > 2,101) \rightarrow$ reject H_0

$\Rightarrow$ We reject the hypothesis that the insurance increases by $ 5000 per $1000 income.

e) Hypotheses :

$$H_0 : B_1 = 1$$

$$H_1 : B_1 > 1$$

Test statistic : $t = \dfrac{b_1 - 1}{se(b_1)} = \dfrac{3,880 - 1}{0,112} = 25,71$

Critical value : $t_{0,99,18} = 2,552$

$\Rightarrow t > 2,552$ $(25,71 > 2,552) \Rightarrow$ reject H_0

$\Rightarrow$ We reject the hypothesis that income increases the amount of life insurance held increase by the same amount, and conclude that the income increases the amount of life insurance held higher than the amount itself.